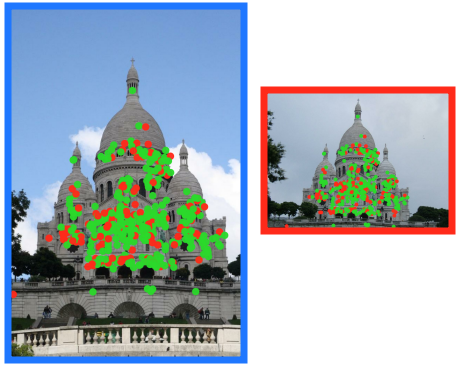
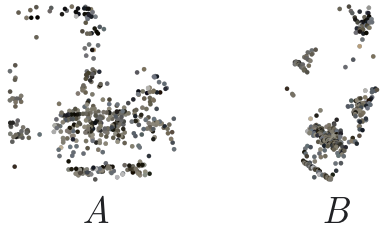
1 Two view subsets



V_A and V_B : disjoint photos of one scene (10 in all). Dots: the 565 points seen at least twice in each subset; red: the 170 points jittered for the test in panel 4.

2 Two reconstructions

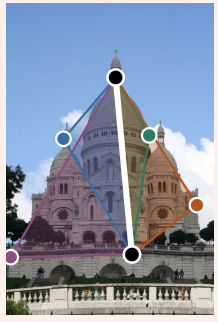


Each subset triangulates the same points on its own. The coordinates differ by an unknown frame and a scale per point:

$$p_k \mapsto \lambda_k H\ p_k \quad (H \text{ any } 4{\times}4)$$

Colours: the photo colour of each point. Same points, so same colours.

3 Frame-free numbers

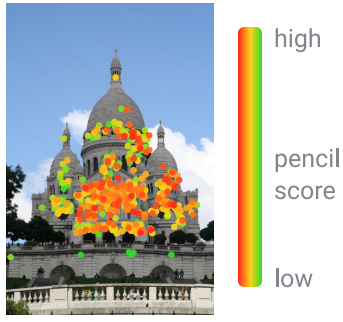


Four planes through the line $p_i p_j$ (white) have a cross-ratio:

$$\mathrm{CR}_{ij} = \frac{[ijce]\ [ijdf]}{[ijcf]\ [ijde]}$$

with brackets $[ijkl] = \det(p_i, \ldots)$. Every frame and every point scale cancels. n points carry $3n - 15$ independent ones.

4 Consistency test



$$S = Q_{0.1}(|\Delta \log|\mathrm{CR}_t||)$$

consistent $S = 0.029$

jittered $S = 0.066$

Jittered: 170 points moved by 8 px in V_B. No frame, camera or noise level is used. 49,991 pencils.